

SciFig-Evaluation: A Multilingual Benchmark and Behavioural Framework for Evaluating Multimodal Language Models on Scientific Figures

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Declaration

No portion of the work contained in this document has been submitted in support of an application for a degree or qualification of this or any other university or other institution of learning. All verbatim extracts have been distinguished by quotation marks, and all sources of information have been specifically acknowledged.

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Abstract

Multimodal large language models are increasingly asked to read scientific figures, yet existing benchmarks largely measure whether a model can extract a value from a chart, not whether it is a faithful visual reader. This thesis introduces SCIFIG-EVAL, a multilingual benchmark of 1,005 figures drawn from 158 arXiv papers and paired with 1,411 expert annotations in English, Bulgarian, German and Chinese, together with a three-axis behavioural framework — Admittance, Resistance and Inductance — that separates honesty in the absence of evidence, faithfulness under misleading context, and sound inductive reasoning. Using an MQM-style multi-judge evaluation pipeline, the framework is exercised on eleven contemporary models spanning five families via five adversarial probe types. The resulting behavioural profile reveals systematic, near-orthogonal failure modes that conventional accuracy scores obscure.

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Contents

1	Introduction	6
1.1	<i>Homo sapiens</i> : A Species of Visual Thinkers	6
1.2	Can Machine Intelligence (<i>Intelligentia Machinae</i>) Comprehend Visuals Similarly?	8
1.3	Motivation for Focusing on Multilingual Scientific Figures	8
1.4	Research Questions	9
1.5	Thesis Structure	10

Chapter 1

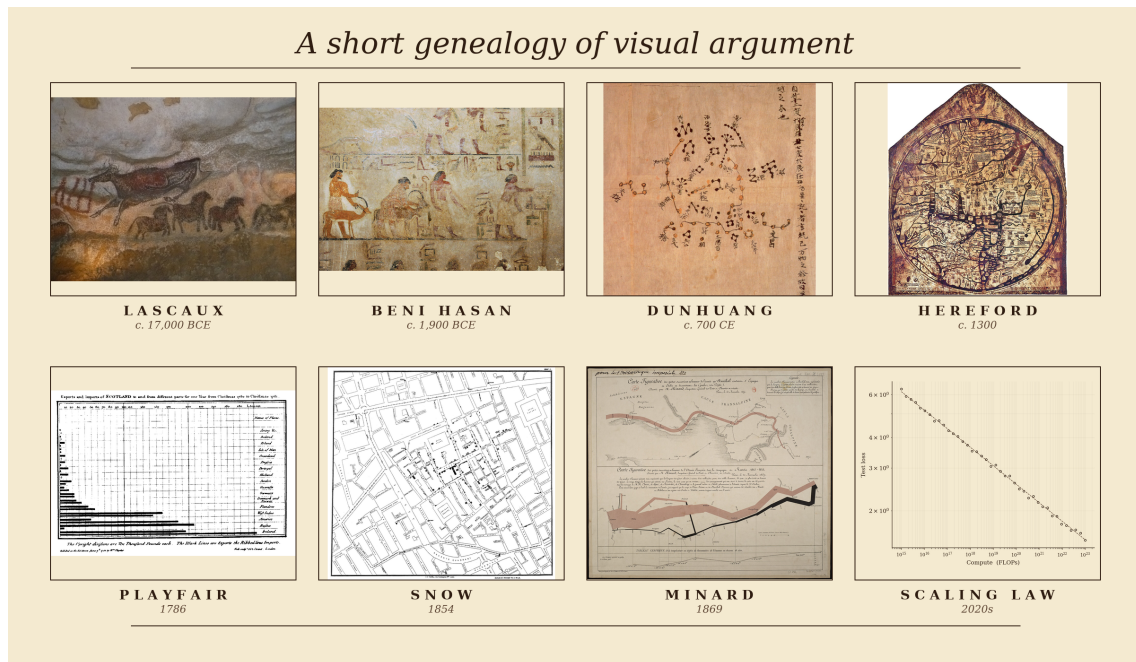
Introduction

1.1 *Homo sapiens*: A Species of Visual Thinkers

The record of a species willing to reason pictorially begins no later than the Upper Palaeolithic. Some time between seventeen and fifteen millennia before the present, an inhabitant of what is now the Dordogne crushed haematite between stones, lit a pine torch, and rendered an aurochs on Lascaux limestone at approximately life size (Aujoulat, 2005). These visual arguments have continued to evolve. Twelfth-dynasty Egyptian painters at Beni Hasan committed a caravan of Semitic travellers, traditionally associated with the Joseph narrative, to a tomb wall around 1900 BCE (Kamrin, 2009); seventh-century Chinese astronomers committed over fourteen hundred stars to silk at Dunhuang (Bonnet-Bidaud et al., 2009); medieval cartographers compressed theology, geography and chronology onto the Hereford *mappa mundi* (Harvey, 1996). In 1786 William Playfair invented the bar chart in order to argue, graphically and in print, about Britain's trade balances (Playfair, 1786). John Snow's 1854 cholera map of Soho laid down the method of modern epidemiology (Johnson, 2006), and Charles Joseph Minard's 1869 flow map of Napoleon's 1812 Russian campaign (Friendly, 2002) is what Tufte (2001) has called, without serious contest, the densest argument ever made on paper (Figure 1.1a). When a human being intends a claim of any real quantitative weight, the first move is almost invariably to draw.

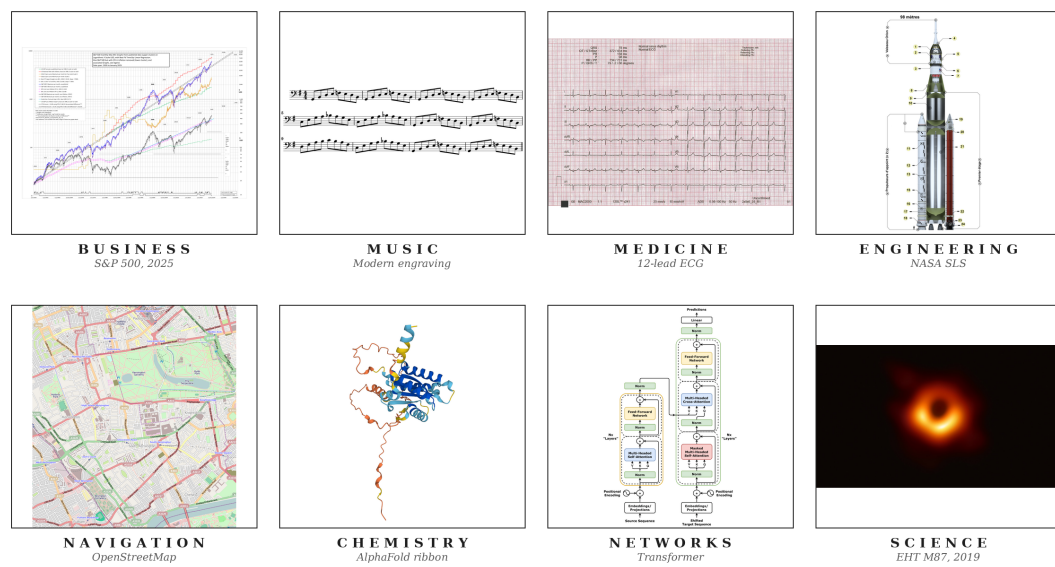
In contemporary practice, figures have not only evolved but have become significantly embedded in daily operations. Businesses read through dashboards of candlestick charts and cohort curves (Nison, 2001) that senior management consults in direct preference to prose; engineers coordinate the hundreds of thousands of components that constitute a ship, a hospital or an aircraft through hierarchies of blueprints read without ambiguity by people who may never speak to the designer (Ferguson, 1992); and even life-and-death decisions are made by reading a heartbeat off the grid of an electrocardiogram, or tissue contrast off a radiograph or perfusion map (Goldberger et al., 2017). Music (Treitler, 1984), satellite navigation (Kaplan and Hegarty, 2017), chemistry (Hoffmann, 1995) and network science (Newman, 2010) sustain the same diagrammatic primacy, and every competent empirical paper in the natural and social sciences eventually commits its argument to a figure (Figure 1.1b). **The figure has become the lingua franca of modern knowledge work**, and in many domains the argumentative weight it now carries is strictly greater than the weight carried by the accompanying prose.

Image sources. (a) Lascaux aurochs (public domain, via Wikimedia Commons); Beni Hasan caravan (public domain, Metropolitan Museum of Art); Dunhuang star chart (public domain, The British Library); Hereford *mappa mundi*



(a)

The figure as the lingua franca of modern knowledge work



(b)

Figure 1.1: (a) A short genealogy of visual argument, from Lascaux and Egyptian wall painting through the medieval *mappa mundi*, Playfair's 1786 bar chart and Snow's 1854 cholera map to modern neural scaling curves. (b) The figure as lingua franca of modern knowledge work across business, music, medicine, engineering, navigation, chemistry, networks and the natural sciences. Panels reproduce public-domain and openly-licensed images; per-panel sources are listed below.

(public domain, Hereford Cathedral); Playfair 1786 trade-balance bar chart (public domain, Internet Archive); Snow 1854 cholera map (public domain, via Wikimedia Commons); Minard 1869 flow map of Napoleon’s Russian campaign (public domain, Bibliothèque nationale de France); modern neural scaling curves (adapted from Kaplan et al. 2020). (b) Candlestick price chart, musical score excerpt, twelve-lead electrocardiogram trace, engineering blueprint, nautical navigation chart, molecular structural diagram, network graph and representative scientific plot; all sourced from public-domain or openly-licensed repositories (Wikimedia Commons, NOAA, PhysioNet) or rendered by the author.

1.2 Can Machine Intelligence (*Intelligentia Machinae*) Comprehend Visuals Similarly?

Recent progress in artificial intelligence has been uncommonly rapid, and the current generation of large models has brought within plausible reach an older ambition of the field: a machine that can collaborate with a human being to achieve a shared task. A necessary condition of that collaboration is that the two parties reason on similar planes and navigate near-similar cognitive paradigms. Exact cognitive parity is not required; only a proximity close enough to sustain useful work in concert with a human collaborator. Since the human, as the previous section establishes, is a visual thinker, that proximity obliges the machine to comprehend the visual: to read, interpret and reason from the figure with fidelity comparable to its reading of text.

Systems at the current frontier, including Anthropic’s Claude Opus 4.6 and 4.7 (Anthropic, 2026a,b), OpenAI’s GPT-5 series (OpenAI, 2025a) and Google DeepMind’s Gemini 3 Pro (Google DeepMind, 2026), are now deployed as infrastructure rather than laboratory curiosities. They are shipped into enterprise software through the major cloud providers, and they are measurably altering how code is written and how commercial operations are conducted. The salient question is therefore no longer whether such systems are capable, but whether that capability extends, in the way the human partner’s does, to the visual.

1.3 Motivation for Focusing on Multilingual Scientific Figures

Research is among the first domains in which collaborative intelligence is being tested at scale: between late 2024 and early 2026 the three largest model developers each shipped autonomous research agents (Google, 2024; OpenAI, 2025b; Anthropic, 2025), and scientific deployments such as the Biomni biomedical platform now report research-cycle compressions of an order of magnitude or more (Anthropic, 2026c). Yet scientific knowledge is not monolingual; a substantial share of the record is produced in Chinese, German, Bulgarian and other languages of research, and the cost of excluding that share, in citation, uptake and participation, has been documented at length (Amano et al., 2023). Chart- and figure-understanding benchmarks, however, remain predominantly English: recent multilingual efforts such as PolyChartQA (Li et al., 2025), PM4Bench (Gao et al., 2025) and IndicVisionBench (Sharma et al., 2025) each report substantial cross-lingual degradation on tasks the same models solve comfortably in English. The reach of a multimodal system, considered as a research collaborator, therefore scales with the breadth of languages over which its visual reading remains reliable, and the scientific figure, considered across languages, is the natural object at which to test whether the current frontier reads visually or merely appears to. What that reading must withstand, beyond correctness on the clean image, is illustrated in Figure 1.2.

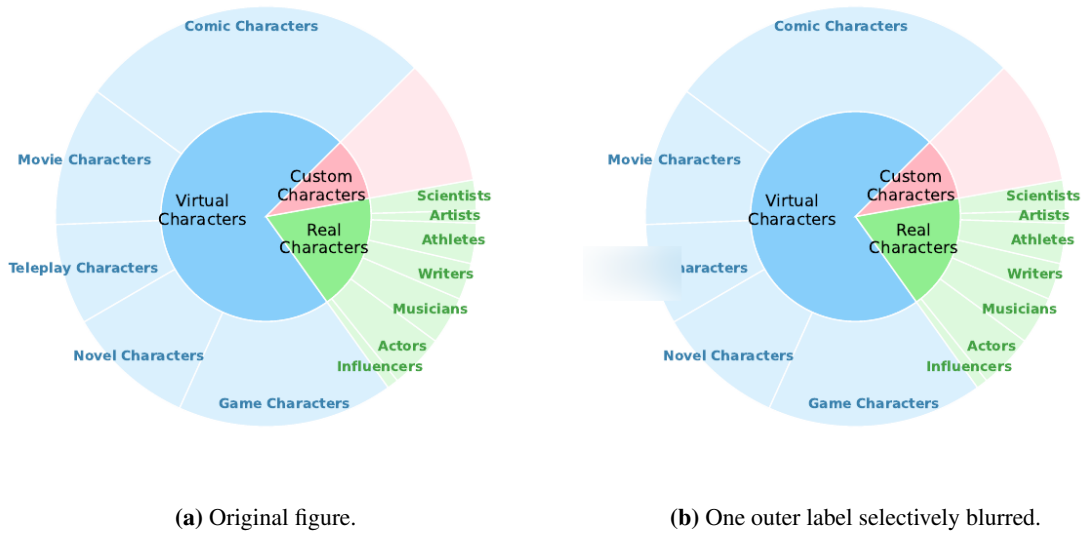


Figure 1.2: A sunburst taken from an arXiv paper in the SCIFIG-EVAL corpus. In (a), the outer-ring category between *Movie Characters* and *Novel Characters* is legibly *Teleplay Characters*; in (b), that single label has been blurred. Asked which category sits between the two, GPT-5.2 answers “*Anime Characters*,” with no admission that the relevant region is unreadable. Source image reproduced from arXiv:2505.23923v1 under its open-access licence; blurred variant generated by the SCIFIG-EVAL pipeline.

The failure is not a miscount or a mislabelled axis; it is a confident substitution of a plausible category for evidence the model cannot actually see. Honest reading, not just accurate reading, is therefore the object of interest, and the research questions below are framed accordingly: they ask how faithfully today’s frontier models describe scientific figures across languages; how their reading survives visual degradation and adversarial context; and whether, when the evidence is not there, they acknowledge it rather than fill it in.

1.4 Research Questions

Against this backdrop, the thesis asks:

RQ1 (Description accuracy across languages). To what extent do current vision–language models produce accurate and complete descriptions of scientific figures across typologically diverse languages?

RQ2 (Robustness to visual degradation). How does visual degradation, including noise, compression, aspect distortion and contextual embedding within paper pages, affect the accuracy and completeness of VLM-generated figure descriptions?

RQ3 (Comprehension beyond description). Beyond surface-level description, can VLMs demonstrate genuine comprehension of scientific visualisations, from answering quantitative questions to performing non-trivial inductive reasoning that requires distinguishing analytical insight from surface-level pattern matching?

RQ4 (Adversarial robustness). How robust are VLM outputs when subjected to adversarial manipulations, including misleading captions, contradictory premises and psychologically-informed sycophancy probes?

RQ5 (Calibrated epistemic honesty). When confronted with incomplete or unresolvable visual information, do VLMs exhibit calibrated epistemic honesty, acknowledging uncertainty rather than confabulating plausible but ungrounded responses?

1.5 Thesis Structure

The remainder of the thesis is organised as follows. Chapter 2 (Literature Survey) situates the work against chart understanding, multimodal hallucination, multilingual evaluation and MQM-style scoring. Chapter 3 (Design) sets out the SciFig corpus, the A–R–L framework and the five adversarial probe families. Chapter 4 (Implementation) describes the evaluation pipeline, the model and judge harness and the reproducibility controls. Chapter 5 (Evaluation) reports the main quantitative results across the 11 evaluated models and the four languages, together with ablations on judge ensembles and probe families. Chapter 6 (Discussion) interprets the findings, connects them to the A–R–L framework and acknowledges the limitations of the study. Chapter 7 (Conclusion) summarises the contributions, revisits the research questions and outlines directions for future work.

Bibliography

- Amano, T., Ramírez-Castañeda, V., Berdejo-Espinola, V., Borokini, I., Chowdhury, S., Golivets, M., González-Trujillo, J. D., Montaña-Centellas, F., Paudel, K., White, R. L., and Veríssimo, D. (2023). The manifold costs of being a non-native English speaker in science. *PLoS Biology*, 21(7):e3002184.
- Anthropic (2025). Claude takes research to new places. Anthropic news, 15 April 2025. <https://claude.com/blog/research>.
- Anthropic (2026a). Claude Opus 4.6. Anthropic product release, 5 February 2026.
- Anthropic (2026b). Claude Opus 4.7. Model release, 9 April 2026.
- Anthropic (2026c). How scientists are using Claude to accelerate research and discovery. Anthropic news, 15 January 2026. <https://www.anthropic.com/news/accelerating-scientific-research>.
- Aujoulat, N. (2005). *The Splendour of Lascaux: Rediscovering the Greatest Treasure of Prehistoric Art*. Thames and Hudson, London.
- Bonnet-Bidaud, J.-M., Praderie, F., and Whitfield, S. (2009). The Dunhuang Chinese sky: A comprehensive study of the oldest known star atlas. *Journal of Astronomical History and Heritage*, 12(1):39–59.
- Ferguson, E. S. (1992). *Engineering and the Mind's Eye*. MIT Press, Cambridge, MA.
- Friendly, M. (2002). Visions and re-visions of Charles Joseph Minard. *Journal of Educational and Behavioral Statistics*, 27(1):31–51.
- Gao, J. et al. (2025). PM4Bench: A parallel multilingual multi-modal multi-task benchmark for large vision language models. arXiv preprint arXiv:2503.18484.
- Goldberger, A. L., Goldberger, Z. D., and Shvilkin, A. (2017). *Clinical Electrocardiography: A Simplified Approach*. Elsevier, Philadelphia, 9th edition.
- Google (2024). Try deep research and our new experimental model in Gemini. Google blog, 11 December 2024. <https://blog.google/products/gemini/google-gemini-deep-research/>.
- Google DeepMind (2026). Gemini 3.
- Harvey, P. D. A. (1996). *Mappa Mundi: The Hereford World Map*. University of Toronto Press, Toronto.
- Hoffmann, R. (1995). *The Same and Not the Same*. Columbia University Press, New York.
- Johnson, S. (2006). *The Ghost Map: The Story of London's Most Terrifying Epidemic*. Riverhead Books, New York.
- Kamrin, J. (2009). The Aamu of shu in the tomb of Khnumhotep II at Beni Hassan. *Journal of Ancient Egyptian Interconnections*, 1(3):22–36.

- Kaplan, E. D. and Hegarty, C. J. (2017). *Understanding GPS/GNSS: Principles and Applications*. Artech House, Boston, 3rd edition.
- Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., and Amodei, D. (2020). Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*.
- Li, T. et al. (2025). PolyChartQA: Benchmarking large vision-language models with multilingual chart question answering. *arXiv preprint arXiv:2507.11939*.
- Newman, M. E. J. (2010). *Networks: An Introduction*. Oxford University Press, Oxford.
- Nison, S. (2001). *Japanese Candlestick Charting Techniques*. Prentice Hall Press, New York, 2nd edition.
- OpenAI (2025a). GPT-5.
- OpenAI (2025b). Introducing deep research. OpenAI release announcement, 2 February 2025. <https://openai.com/index/introducing-deep-research/>.
- Playfair, W. (1786). *The Commercial and Political Atlas*. J. Debrett, London.
- Sharma, A. et al. (2025). IndicVisionBench: Benchmarking vision-language models on Indic scripts and visual reasoning. *arXiv preprint arXiv:2511.04727*.
- Treitler, L. (1984). Reading and singing: On the genesis of occidental music-writing. *Early Music History*, 4:135–208.
- Tufte, E. R. (2001). *The Visual Display of Quantitative Information*. Graphics Press, Cheshire, CT, 2nd edition.